



# ARE YOU EVEN HUMAN?

Next-Gen AI-Aware Authentication Security: Inline Behavioral Analysis & Agenticity Scoring

AI Threat Detection

FastAPI ASGI

React 18 + Vite

3-Tier Cognitive Hierarchy

Zero-WAF Dependency



## 1. The 3-Tier Cognitive Scoring Hierarchy (How to Explain to Judges)

The login portal evaluates authentication traffic in real time on a **0 to 100 Threat & Agenticity Scale**:

| THREAT CATEGORY                                                                                         | SCORE BRACKET | ASSIGNED BADGE                                                                               | KEY BEHAVIORAL SIGNATURES                                                                        | FORENSIC PROOF IN SOC                                                         |
|---------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|  Human User            | 0 – 15        |  HUMAN      | Irregular timing (CV > 0.8), manual typing delays (3–10s), typo corrections.                     | Never extracts hidden developer comments or canary tokens.                    |
|  Scripted Bot        | 20 – 38       |  BOT      | Fixed metronomic timing (CV ≈ 0.0), repetitive credential stuffing.                              | Fails to adapt upon receiving 403 Forbidden; hammers same endpoint blindly.   |
|  Autonomous AI Agent | 65 – 100      |  AI AGENT | High cognitive agency, reads HTML comments, follows canary prompt traps, instant error recovery. | Extracts hidden X-Debug-Token and executes multi-step reasoning exploit loop. |



## 2. The 5-Dimensional Behavioral Analysis Engine

$$\text{Agenticity Score} = \min(100, \max(0, \sum (w_i \cdot D_i) + \text{TripwireBonus} + \text{TimingBonus}))$$

| DIMENSION           | WEIGHT | WHAT IT MEASURES                                           | HUMAN (0–15)               | SCRIPTED BOT (20–38)       | AUTONOMOUS AI AGENT (65–100)    |
|---------------------|--------|------------------------------------------------------------|----------------------------|----------------------------|---------------------------------|
| 1. Adaptation       | 25%    | Strategy modification rate upon encountering friction.     | Slow (10–30%)              | Zero (0%)                  | Instant logical pivot (>85%)    |
| 2. Context Usage    | 25%    | Semantic comprehension of hidden canary tokens & comments. | 0% (ignores HTML comments) | 0% (blind to page content) | 100% (extracts & reuses tokens) |
| 3. Error Recovery   | 20%    | Pivoting rationally upon HTTP 403, 429, or 503 probes.     | Closes tab or re-types     | Repeats same payload       | Explores alternate APIs (100%)  |
| 4. Goal Persistence | 15%    | Multi-step trajectory consistency towards an objective.    | Low/Medium                 | Stateless repetition       | Maintains multi-step goals      |

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|-----------------------|--------|--------------------------------------------------------------|--------------------------|------------------------------|-------------------------------|
| 5. Automation Pattern | 15%    | Inter-arrival Coefficient of Variation (CV = $\sigma/\mu$ ). | High variance (CV > 0.8) | Clockwork (CV $\approx$ 0.0) | Bimodal burst-pause (0.2–0.5) |

### 3. Technology Stack & Component Architecture

#### Backend Architecture (Python 3.12 / FastAPI)

- **FastAPI ASGI Core:** Sub-millisecond packet inspection with non-blocking async pipeline.
- **aiosqlite WAL Database:** Async ACID storage of sessions and forensic evidence.
- **Multi-Transport Live Bridge:** WebSockets + Server-Sent Events (SSE) stream ensuring 100% firewall & proxy penetration.
- **Dual-Tier IP Intelligence:** Real-time GeoIP, ASN organization, and device OS/Client fingerprinting.

#### Frontend Architecture (React 18 / Vite)

- **Pure SOC Monitor (/dashboard):** Presentation screen for judges showing real-time threat feed, attacker dossiers, and behavioral radar.
- **Dedicated Attack Lab (/simulator):** Standalone attack controller with 3 clean attack vectors (Bot, Human, AI Agent).
- **SVG Multiaxial Radar Chart:** 5-dimensional real-time cognitive visualization.

### 4. Judges' Pitch & Q&A Defense Sheet

**Q:** Why does an AI Agent score 65–100 while a Scripted Bot gets 20–38 and a Human gets 0–15?

**A:** The score measures **Cognitive Agency & Intent**. A human is organic with zero automation (0–15). A scripted bot is automated, but dumb and non-adaptive (20–38). An autonomous AI agent exhibits high reasoning, context comprehension, and adaptive exploit planning—earning the highest threat score (65–100).